KERALA PUBLIC SERVICE COMMISSION • OFFICIAL SYLLABUS NOTE

SURVEYOR GRADE II (SURVEY & LAND RECORDS) & KWA TRACER

Cat. Nos: 411/2022, 692/2022, 106/2017, 043/2023 • Technical Education & Town Planning

MODULE 09

Workshop Calculation and Science

Official Weightage: 10 Marks Marks

PRESCRIBED SYLLABUS TOPICS COVERED:

Units & Measurements (SI Units, Metric & Imperial conversions: Acres, Hectares, Cents, Square feet), Arithmetic (Fractions, Square roots, Ratio & Proportion, Percentages), Mensuration of 2D Figures (Triangles, Trapezium, Circles, Sectors, Ellipse), Mensuration of 3D Solids (Cubes, Cylinders, Cones, Spheres, Frustums), Earthwork Volume Computation (Prismoidal Formula & End-Area rule), Mass, Weight, Density & Specific Gravity, Speed, Velocity & Newton's Laws, Work, Power & Energy (Metric & British Horsepower), Simple Machines (MA, VR, Efficiency, Levers), Heat & Temperature Conversions, Basic Trigonometry & Heights/Distances.

+P COMPREHENSIVE EXAM CRACKER FEATURES

- Strictly aligned with Kerala PSC latest question patterns (Land Records & KWA)
- Contains verified formulas, numerical shortcuts, standard tables & specifications
- Highlights repeatedly tested PSC trap questions & previous year keys

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CHAPTER 1: UNITS, MEASUREMENTS & CONVERSIONS

Technical calculations in surveying and drafting require absolute fluency in the International System of Units (SI), metric units, and traditional land measurement conversions used in Kerala revenue records.

1.1 Fundamental SI Base Units & Derived Units

Physical Quantity	SI Unit Name	Symbol	Definition / Derivation Formula
Length	Metre	m	Base SI unit of linear distance
Mass	Kilogram	kg	Base SI unit of quantity of matter
Time	Second	s	Base SI unit of duration
Force	Newton	N	1 N = 1 kg·m/s² (Force giving 1 kg mass an acceleration of 1 m/s²)
Pressure / Stress	Pascal	Pa	1 Pa = 1 N/m²; 1 bar = 10 u Pa; 1 N/mm² = 1 MPa = 10 v Pa
Work / Energy	Joule	J	1 J = 1 N·m (Work done when 1 N force moves through 1 m distance)
Power	Watt	W	1 W = 1 J/s (Rate of doing work at 1 Joule per second)

1.2 Land Measurement Conversion Factors (Kerala Cadastral Standards)

Unit of Area	Equivalent Metric / Imperial Value	Kerala Land Records Relation
1 Cent	40.4686 m² "H 40.47 m² = 435.6 sq ft	Standard land parcel subdivision unit in Kerala
1 Acre	100 Cents = 4,046.86 m² = 43,560 sq ft = 0.4047 hectare	Standard agricultural land unit; 10 sq Gunter's chains
1 Hectare (ha)	10,000 m² = 2.471 Acres = 247.1 Cents	Standard metric forestry and large estate unit
1 Are	100 m² = 2.471 Cents	Metric subdivision unit (100 Ares = 1 Hectare)
1 Square Metre (m²)	10.764 sq ft = 0.0247 Cents	Standard engineering floor area unit
1 Square Foot (sq ft)	0.0929 m² = 144 sq inches	Building plinth area standard

+P PSC FOCUS POINT: KERALA REVENUE LAND CONVERSION TRIPLETS

- 1 Acre = 100 Cents = 43,560 sq ft = 4,046.86 m².
- 1 Cent = 435.6 sq ft = 40.47 m².
- 1 Hectare = 10,000 m² = 2.471 Acres = 247.1 Cents.
- 1 Are = 100 m² = 2.471 Cents.

CHAPTER 2: APPLIED ARITHMETIC, RATIOS & PERCENTAGES

Foundational arithmetic concepts regularly tested in technical examinations.

2.1 Ratio & Proportion

• **Direct Proportion:** When two quantities increase or decrease together in the same ratio: x • / y • = x , / y , (e.g. chain length and true distance, quantity of bricks and wall volume).

• **Inverse Proportion:** When an increase in one quantity causes a proportional decrease in the other: x • · y • = x , · y , (e.g. number of workers and days to complete earthwork excavation, scale denominator and map feature size).

• **Rule of Three:** Given three terms of a proportion a : b = c : x, finding the fourth term: x = (b · c) / a.

2.2 Percentages & Profit/Loss

Ø=ÜÐ CORE FORMULA VAULT: PERCENTAGE FORMULAS

Percentage Value = (Part Value / Total Base Value) × 100%

Percentage Error = (Measured Value - True Value) / True Value × 100%

Quantity after p% Increase = Original × (1 + p / 100)

Quantity after p% Decrease = Original × (1 - p / 100)

CHAPTER 3: MENSURATION: 2D PLANE FIGURES

Mensuration deals with calculating perimeters and areas of geometric surfaces.

3.1 Triangles

Triangle Type	Area Formula	Perimeter Formula & Notes
Right-Angled Triangle	Area = 1/2 × Base × Height = 1/2 · b · h	Perimeter P = b + h + " (b² + h²)
Scalene Triangle (Heron's Formula)	Area = " [s · (s - a) · (s - b) · (s - c)]	Semi-perimeter s = (a + b + c) / 2
Equilateral Triangle (Side a)	Area = (" 3 / 4) · a² "H 0.433 · a²	Height h = (" 3 / 2) · a; Perimeter P = 3a
Isosceles Triangle (Equal sides a, Base b)	Area = (b / 4) · " (4a² - b²)	Height h = " (a² - b²/4); Perimeter P = 2a + b

3.2 Quadrilaterals & Polygons

Plane Figure	Area Formula	Perimeter Formula & Geometric Notes
Rectangle (Sides a, b)	Area = a · b	P = 2(a + b); Diagonal d = " (a² + b²)
Square (Side a)	Area = a² = 1/2 · d²	P = 4a; Diagonal d = a" 2
Parallelogram (Base b, Height h)	Area = Base × Height = b · h	P = 2(a + b); Adjacent sides a, b
Rhombus (Diagonals d •, d ,)	Area = 1/2 · d • · d ,	P = 4a; Diagonals bisect at right angles (90°)
Trapezium (Parallel sides a, b; Height h)	Area = 1/2 · (a + b) · h	P = a + b + c + d; Height is perpendicular distance between parallel sides
Regular Hexagon (Side a)	Area = (3" 3 / 2) · a² "H 2.598 · a²	Composed of 6 equilateral triangles; P = 6a

3.3 Circles, Sectors & Ellipse

Curved Figure	Area Formula	Perimeter / Arc Length Formula
Circle (Radius r, Diameter d)	Area = < +" Ò ƒÀ / 4) · d²	Circumference C = 2< " Ò Àd
Semi-Circle	Area = 1/2 · < +	Perimeter P = < " ² "' Ò "ƒÀ + 2) "H (36 / 7) · r
Circular Ring (Outer R, Inner r)	Area = < …+" Ò +"' Ò À(R + r)(R - r)	Width of ring w = R - r
Sector of Circle (Angle ;‹ •	Area = (;, ò 3c ’ r Àr² = 1/2 · l · r	Arc length l = (;, ò 3c ’ r #Àr; Perimeter P = l + 2r
Ellipse (Semi-axes a, b)	Area = < r r	Perimeter P "H < ² 2† ² "' Ò !¢,ƒ6 ² "'† ² 6"'' Ð

+P PSC FOCUS POINT: PERIMETER OF SEMI-CIRCLE PSC TRAP

PSC Question: 'What is the perimeter of a semi-circle of radius r?'

Perimeter = Curved arc + Diameter = < " ² ‴ Ò "ƒ À + 2) = r(22/7 + 2) = (36 / 7) · r.

Never write 1/2 × 2< " Ò Àr (that is only the curved arc, omitting the closing base diameter!).

CHAPTER 4: MENSURATION: 3D SOLID GEOMETRY

Calculating volumes, curved surface areas, and total surface areas of engineering solids.

Solid	Volume (V)	Curved / Lateral Surface Area	Total Surface Area (TSA)
Cube (Side a)	V = a^3	Lateral Area = $4a^2$	TSA = $6a^2$; Solid Diagonal = a" 3
Cuboid (l, b, h)	V = l · b · h	Lateral Area = 2(l + b) · h	TSA = 2(lb + bh + hl); Diagonal = " ($l^2 + b^2 + h^2$)
Cylinder (Radius r, Height h)	V = < +&€	Curved Area = 2< &€	TSA = 2< &, ² #Àr² = 2< "‡" ² ,•
Hollow Cylinder (R, r, h)	V = < …+" Ò +‴ r €	CSA = 2< ,…" ² "•	TSA = 2< ,…" ² ‴ ² #À(R^2 - r^2)
Cone (Radius r, Height h)	V = 1/3 · < +&€	Curved Area = < l (Slant height l = " ($r^2 + h^2$))	TSA = < l + < +" Ò Àr(l + r)
Sphere (Radius r)	V = 4/3 · < +0	Surface Area = 4< +	Surface Area = 4< +
Hemisphere (Radius r)	V = 2/3 · < +0	Curved Area = 2< +	TSA = 2< +" ² Àr² = 3< +
Frustum of Cone (R, r, h)	V = 1/3 · < , r …+" ² +" ² ‴•	CSA = < Â…" ² ‴ …6Æ çB Â Ò !¢†‹"₹SA"Ð r)²))	CSA + < +" ² Àr²

CHAPTER 5: EARTHWORK & VOLUME COMPUTATION RULES

Calculating excavation and embankment volumes for civil engineering roads, canals, railways, and reservoirs.

5.1 Prismoidal Formula vs Average End Area Rule

Method	Governing Mathematical Formula	Assumed Boundary / Nature	
Average End Area (Trapezoidal) Rule	Volume V = d · [(A • + An) / 2 + A , + A ƒ + ... + An-1] where d = common distance between cross-sections	Assumes longitudinal slope between sections is straight planar surface	Any number o
Prismoidal Formula (Simpson's Rule for Volume)	Volume V = (d / 3) · [(A • + An) + 4 · (:2 W`en Areas) + 2 · (:2 öFB &V 2' Ð	Assumes surface between sections is parabolic or ruled prismoidal surface; mathematically exact for prismoids	Strictly require

Ø=ÜÐ CORE FORMULA VAULT: PRISMOIDAL CORRECTION FORMULA

V_pris = V_trap - Cp

For a prismoid of length L with side slope s:1 and centerline depths d •, d ,:

Cp = (L / 6) × s × (d • - d ,)²

Notice: Prismoidal correction is ALWAYS SUBTRACTIVE (-), because (d • - d ,)² is always positive!

CHAPTER 6: APPLIED MECHANICS: MASS, FORCE & MOTION

Fundamental physical principles governing materials, loads, and vehicles in survey academy science.

6.1 Mass, Density & Specific Gravity

• **Mass (m):** The quantity of matter contained in a body. Constant everywhere in the universe (SI unit: kg).

• **Weight (W):** The gravitational force exerted by the Earth on a body: W = m · g (SI unit: Newton N). Varies with local acceleration due to gravity (g "H 9.81 m/s²).

• **Density (< ‰:** Mass per unit volume of a substance: < = Mass / Volume (kg/m³). Density of pure water at 4°C = 1,000 kg/m³ = 1.0 g/cm³.

• **Specific Gravity (Relative Density):** The dimensionless ratio of the density of a substance to the density of pure water at 4°C: Sp. Gr. = < ÷7V'7F æ6R ò Á_water. Pure water = 1.0; Cement = 3.15; Steel = 7.85; Mercury = 13.6.

6.2 Speed, Velocity & Equations of Uniform Motion

Equation of Motion	Variables Involved	Application Condition
v = u + a · t	u = initial velocity, v = final velocity, a = acceleration, t = time	Uniform acceleration in a straight line
s = u · t + 1/2 · a · t²	s = displacement distance traversed	Uniform acceleration from initial speed
v² = u² + 2 · a · s	Independent of elapsed time t	Calculating vehicle stopping distance
v_avg = (u + v) / 2	Average velocity during interval	Linear acceleration

6.3 Friction

• **Frictional Force (F):** The tangential resisting force opposing the relative motion between two contacting surfaces: F = μ · N, where μ = coefficient of friction, N = normal reaction force.

• **Angle of Friction (<b‟:** The angle between the resultant reaction and the normal to the contact plane when limiting sliding motion is impending: tan <b Ò Rà

• **Angle of Repose (;,“:** The maximum angle of an inclined plane at which a granular body (earth, sand) rests in equilibrium without sliding down: Angle of Repose ;, Ò ævÆR öb `riction <bà

CHAPTER 7: WORK, POWER, ENERGY & SIMPLE MACHINES

The mechanical transformations of force and displacement.

7.1 Work, Power & Energy Definitions

Ø=ÜÐ CORE FORMULA VAULT: WORK, POWER & ENERGY EQUATIONS

Work Done W = Force × Displacement = F · s · cos ;, ´¦÷VÆW2 ¥Ð

Work Done against Gravity = m · g · h [Joules]

Power P = Work Done / Time = (F · s) / t = F · v [Watts W]

Horsepower (HP) Conversions:

• 1 Metric Horsepower (PS) = 75 kg·m/s = 735.5 Watts

• 1 British Horsepower (HP) = 550 ft·lb/s = 746.0 Watts

Mechanical Energy:

• Kinetic Energy KE = 1/2 · m · v² [Joules]

• Potential Energy PE = m · g · h [Joules]

+P PSC FOCUS POINT: METRIC VS BRITISH HORSEPOWER IN PSC

• 1 Metric HP = 735.5 Watts (75 kg·m/s).

• 1 British HP = 746 Watts (550 ft·lb/s).

PSC questions regularly ask for these exact numeric constants!

7.2 Simple Machines: MA, VR & Efficiency

Parameter	Rigorous Engineering Definition	Governing Formula
Mechanical Advantage (MA)	Ratio of resistance load lifted (W) to effort force applied (P)	MA = Load / Effort = W / P (dimensionless)
Velocity Ratio (VR)	Ratio of distance moved by effort (y) to distance moved by load (x)	VR = Distance of Effort / Distance of Load = y / x (purely geometric; constant for machine)
Efficiency (;r•	Ratio of useful work output to total work input	Efficiency ;r Ò „Ô ò VR) × 100%
Ideal (Frictionless) Machine	Machine having 100% efficiency (no friction losses)	For ideal machine: Efficiency ;r Ò R => MA = VR

7.3 Levers: The Three Classes

• **First Class Lever:** Fulcrum (F) is in the MIDDLE between Effort (E) and Load (L): E - F - L. Examples: Crowbar, scissors, seesaw, pair of pliers. MA can be > 1, = 1, or < 1.

• **Second Class Lever:** Load (L) is in the MIDDLE between Fulcrum (F) and Effort (E): F - L - E. Examples: Wheelbarrow, nutcracker, paper cutter. MA is ALWAYS GREATER THAN 1 (MA > 1)!

• **Third Class Lever:** Effort (E) is in the MIDDLE between Fulcrum (F) and Load (L): F - E - L. Examples: Pair of tongs, human forearm, tweezers, shovel. MA is ALWAYS LESS THAN 1 (MA < 1)!

CHAPTER 8: HEAT, TEMPERATURE & BASIC TRIGONOMETRY

Thermodynamics scales and trigonometric geometry.

8.1 Temperature Scales Inter-Conversion

Ø=ÜÐ CORE FORMULA VAULT: TEMPERATURE CONVERSION FORMULAS

Fundamental Relationship: C / 5 = (F - 32) / 9 = (K - 273.15) / 5

• **Celsius to Fahrenheit: F = (9 / 5 × C) + 32 = (1.8 × C) + 32**

• **Fahrenheit to Celsius: C = 5 / 9 × (F - 32)**

• **Celsius to Kelvin: K = C + 273.15**

• **Temperature where Celsius and Fahrenheit scales read identical: - 40° (i.e. - 40°C = - 40°F)!**

8.2 Basic Trigonometry & Identities

Angle (;,•	sin ;€	cos ;€	tan ;€	cot ;€	sec ;€	cosec ;€
0°	0	1	0	Undefined	1	Undefined
30°	1/2 (0.5)	" 3 / 2 (0.866)	1 / " 3 (0.577)	" 3 (1.732)	2 / " 3 (1.155)	2
45°	1 / " 2 (0.707)	1 / " 2 (0.707)	1	1	" 2 (1.414)	" 2 (1.414)
60°	" 3 / 2 (0.866)	1/2 (0.5)	" 3 (1.732)	1 / " 3 (0.577)	2	2 / " 3 (1.155)
90°	1	0	Undefined	0	Undefined	1

8.3 Heights & Distances in Survey Trigonometry

Determining elevations and horizontal clearances using measured angles of elevation or depression is a daily surveyor application.

- **Accessible Base Case:** When distance D to base of vertical tower is accessible and measured: Height h = D · tan ;,à If instrument axis height is i, total elevation H = i + D · tan ;,à
- **Inaccessible Base (Two Instrument Stations):** When base is inaccessible (e.g. across a river): Two stations A and B are established in line with tower at distance b apart. Observing angles of elevation ; at A and ; at B (where ; < ;"“ Height h = (b · sin ; r 6–â ²) / sin(;" Ò ±) = b / (cot ; Ò 6÷B ²).

Ø=ÜÐ CORE FORMULA VAULT: HEIGHTS & DISTANCES FORMULA

Height of Inaccessible Object h = b / (cot ; Ò 6÷B ²)

Horizontal Distance to Object D = b · cot ;" ò †6÷B ± - cot ;"•

where b = horizontal distance between observation stations A and B, ; À ;" Ò ævÆW2 öb VÆPvation

CHAPTER 9: PSC SOLVED NUMERICAL MODELS

Ø=ÜÝ PSC SOLVED NUMERICAL: Land Area Conversion: Square Feet to Cents and Acres

Question: A rectangular residential plot in Kerala measures 66 feet by 132 feet. Calculate its area in: (a) Square feet, (b) Cents, (c) Acres.

Solution:
Area in sq ft = 66 ft × 132 ft = 8,712 sq ft
Rule: 1 Cent = 435.6 sq ft
(b) Area in Cents = 8,712 / 435.6 = 20.0 Cents
Rule: 1 Acre = 100 Cents = 43,560 sq ft
(c) Area in Acres = 20.0 / 100 = 0.20 Acre

Correct Option / Answer: Area = 8,712 sq ft = 20.0 Cents = 0.20 Acre

Ø=ÜÝ PSC SOLVED NUMERICAL: Earthwork Volume by Prismoidal Formula

Question: Three consecutive cross-sectional areas of a railway cutting spaced at 30 m intervals are: A • = 40 m², A , = 65 m², A ƒ = 80 m². Calculate the volume of earthwork using the Prismoidal Formula.

Solution:
Common interval d = 30 m
End areas: A • = 40 m², A ƒ = 80 m² => (A • + A ƒ) = 120 m²
Middle area (even): A , = 65 m²
Formula: V = (d / 3) × [(A • + A ƒ) + 4(A ,)]
V = (30 / 3) × [120 + 4(65)] = 10 × [120 + 260] = 10 × 380 = 3,800 m³

Correct Option / Answer: 3,800 m³

Ø=ÜÝ PSC SOLVED NUMERICAL: Temperature Scale Coincidence

Question: At what temperature do the Celsius and Fahrenheit thermometers indicate the exact same numerical value?

Solution:
Set C = F = x in the formula: C / 5 = (F - 32) / 9
x / 5 = (x - 32) / 9
9x = 5(x - 32) = 5x - 160
9x - 5x = - 160 => 4x = - 160 => x = - 40

Correct Option / Answer: - 40° (i.e. - 40°C = - 40°F)

Ø=ÜÝ PSC SOLVED NUMERICAL: Simple Machine Mechanical Advantage & Efficiency

Question: A simple screw jack with a velocity ratio of 250 is used to lift a car weighing 12 kN (12,000 N) by applying an effort force of 80 N. Calculate the Mechanical Advantage and the Efficiency of the jack.

Solution:
Load W = 12,000 N, Effort P = 80 N, Velocity Ratio VR = 250
Mechanical Advantage MA = W / P = 12,000 / 80 = 150
Efficiency ;r Ò „Ô ò VR) × 100% = (150 / 250) × 100% = 0.60 × 100% = 60%

Correct Option / Answer: MA = 150; Efficiency = 60%

Ø=ÜÝ PSC SOLVED NUMERICAL: Trigonometric Height of Inaccessible Monument

Question: To find the height of a chimney, the theodolite is set up at station A and the angle of elevation to the top is measured as 30°. Moving 50 m closer to the chimney to station B, the angle of elevation is measured as 45°. If the height of the instrument is 1.5 m, find the total height of the chimney above ground.

Solution:
Distance between stations b = 50 m
Angle at A (; ’ Ò 3 Óâ 6÷B 3 Ò !£2 $, ãs3
Angle at B (;"’ Ò C[Óâ 6÷B C[Ò ã
Height above instrument h = b / (cot ; Ò 6÷B ²) = 50 / (1.732 - 1.000) = 50 / 0.732 = 68.306 m
Total Height above ground H = h + Instrument Height = 68.306 + 1.500 = 69.806 m "H 69.81 m

Correct Option / Answer: 69.81 metres

CHAPTER 10: KERALA PSC 50 RAPID-FIRE ONE-LINERS & EXAM VAULT

Rapid-Fire PSC One-Liners: Units, Area & Mensuration

- **Number of square feet in 1 Cent:** 435.6 sq ft (40.47 m²).
- **Number of Cents in 1 Acre:** 100 Cents (43,560 sq ft).
- **Number of square metres in 1 Hectare:** 10,000 m² (2.471 Acres).
- **Perimeter of semi-circle of radius r:** r(< ² "’ Ò ƒ3b ò r’ r .
- **Area of equilateral triangle with side a:** (" 3 / 4) · a² "H 0.433 a².
- **Area of trapezium formula:** 1/2 × (Sum of parallel sides) × Height.
- **Volume of cone compared to cylinder of same base & height:** 1/3rd of cylinder volume (V = 1/3 < +&,’à
- **Prismoidal formula for earthwork volume requires:** Strictly an ODD number of cross-sections.
- **Prismoidal correction to trapezoidal volume is always:** Subtractive (Cp = L/6 · s · (d1 - d2)²).
- **Diagonal of a cube of edge a:** a" 3.

Rapid-Fire PSC One-Liners: Mechanics, Work & Power

- **Value of 1 Metric Horsepower in Watts:** 735.5 Watts (75 kg·m/s).
- **Value of 1 British Horsepower in Watts:** 746.0 Watts (550 ft·lb/s).
- **Specific gravity of pure water at 4°C:** 1.0 (Density = 1000 kg/m³).
- **Specific gravity of ordinary Portland cement:** 3.15.
- **Specific gravity of structural steel:** 7.85.
- **Specific gravity of mercury:** 13.6.
- **Formula for kinetic energy:** 1/2 · m · v².
- **In a machine, Efficiency is the ratio of:** Mechanical Advantage to Velocity Ratio (MA / VR).
- **Lever having mechanical advantage always greater than 1:** Second Class Lever (e.g. Wheelbarrow).
- **Lever having mechanical advantage always less than 1:** Third Class Lever (e.g. Tongs, tweezers).

Rapid-Fire PSC One-Liners: Heat, Physics & Trigonometry

- **Temperature at which Celsius and Fahrenheit are equal:** - 40 degrees (- 40°C = - 40°F).
- **Boiling point of water in Fahrenheit scale:** 212°F (Freezing point = 32°F).
- **Absolute zero temperature in Celsius:** - 273.15°C (0 Kelvin).
- **Value of sin 30°:** 1/2 (0.5).
- **Value of tan 45°:** 1.0.
- **Value of cos 60°:** 1/2 (0.5).
- **Value of sin 60°:** " 3 / 2 "H 0.866.
- **Identity: sin² ;, ² 6÷;" ¸ equals** 1.
- **Identity: 1 + tan² ;, W V Ç3** sec² ;,à
- **Relation between angle of repose and angle of friction:** Angle of Repose equals Angle of Friction (;, Ò Æ).

Workshop Calculation Revision Checklist Matrix

Physical Concept	Governing Formula / Conversion	Standard Kerala PSC Key Value
1 Cent of Land	435.6 sq ft = 40.4686 m²	100 Cents = 1 Acre (43,560 sq ft)
1 Hectare	10,000 m² = 2.471 Acres = 247.1 Cents	1 Are = 100 m² = 2.471 Cents
Horsepower	1 Metric HP = 735.5 W; 1 British HP = 746 W	75 kg·m/s for Metric HP
Temperature Coincidence	C / 5 = (F - 32) / 9	- 40° is identical on both C and F scales
Machine Efficiency	;r Ò Ô ò VR × 100%	For ideal machine ;r Ò R æB Ô Ò VR
Semi-Circle Perimeter	P = r(< ² "' $, ƒ 3b ò r' r	Includes both curved arc and base diameter
Prismoidal Rule	V = (d / 3) [(A • + An) + 4(Even) + 2(Odd)]	Exact for prismoids; requires odd cross-sections
Prismoidal Correction	Cp = (L / 6) · s · (d • - d ,)²	Always subtractive from Trapezoidal volume